SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15 19

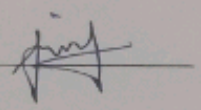
COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5
Duration : 60 Minutes

Code of Conduct

I Jannathul Firdaus. N (192573025) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded

T2421F

1. Compare Type 1 and Type 2 hypervisors and mission - Critical Applications.

A hypervisor is software that creates and manages virtual machines (VM) allowing multiple Operating Systems to run on a single physical server.

Comparison:

Feature	Type 1 Hypervisor	Type 2 hypervisor.
Architecture	run directly on hardware	Run on a host Operating System
Performance	high performance with low overhead	comparitively lower performance
Security	generally smaller attack surface	Depends on the security of the host OS and hypervisor
management	suitable for large scale environments	suitable for personal use and testing.
Example	VMware ESXi, Xen	Virtual Box, VMware workstation.

Analysis:

Type 1 hypervisor provide better performance because they allow hardware directly. they offer efficient resources management, scalability and strong isolation when properly configured.

Type 2 hypervisors are easier to install but depend on the host Operating System, which can introduce additional overhead and security risk.

2. workload Isolation for Finance, Healthcare, and student portals.

Virtualization allows finance, healthcare and student portal applications to run on the same physical hardware while maintaining logical isolation between the workloads.

How virtualization Support isolation:

CPU isolation: The hypervisor allocates CPU resources to individual VMs.

Memory isolation: Memory protection prevents one VM from accessing another VM's memory.

Storage isolation: Separate virtual disks and access permissions protect sensitive data.

Network isolation: Virtual networks and firewall restrict unauthorized communicati

Access control: Role based access control limits access to unauthorized users.

Two Major Attack Surfaces:

Hypervisor attacks: Vulnerabilities in the hypervisor may allow attackers to escape a VM and access other workloads.

Solution: Regularly patch the hypervisor, restrict administrative access and enable security monitoring.

Virtual network attacks: Misconfigured networks may allow unauthorized access between VMs.

solution: Use Network segmentation, firewalls, encryption and strict access policies.

3. Analyze VM host Utilization and Recommend corrective Actions:

Given utilization Data:

Parameter	Value	Analysis
CPU utilization	92%	Very high CPU Usage
Memory utilization	88%	High memory consumption
Storage I/O latency	35 ms	Potential Storage bottleneck.
Avarage boot time	170 sec	Slow VM startup.

Analysis and corrective actions:

CPU bottle neck: CPU utilization of 92% indicates limited processing capacity and possible competition between VMs.

solution: Identify CPU intensive VMs optimize VCPU allocations, and migrate workloads to less-loaded hosts.

MONOMAX T2421F

Memory Bottleneck: Memory utilization of 85% indicates limited available memory and possible memory pressure.

Solution: Optimize VM memory allocation, memory swapping and add RAM if required.

Storage bottleneck: Storage latency of 35ms may indicate slow disk or excessive I/O requests, affecting application and VM performance

Solution: Identify I/O intensive VMs, optimize storage usage and consider faster SSD based storage.

Slow VM boot time: The 120 sec boot may result from resource contention, storage delays or excessive startup services.

Solution: Reduce unnecessary startup services, avoid simultaneous VM startups, and improve storage performance.

4. How Software defined Datacenter (SDDC) Improves Agility:

An SDDC is a Datacenter in which compute, Storage and networking resources are virtualized and managed through software.

Traditional Datacenter Vs. SDDC

Feature	Traditional Datacenter	SDDC
compute	physical server based	virtualized compute resources.
Storage	Dedicated storage hardware	software defined storage pools.
networking	Mostly manual configuration	programmable virtual networks.
provisioning	Slower and manual	Faster and automated
management	hardware-specific tools	centralized software management

T2421F

Key components:

Compute Virtualization: hypervisor divide physical servers into multiple VMs. This improves resource utilization and enables faster applications.

Storage Virtualization: physical storage is combined into logical pools. This simplifies storage allocation and supports features such as snapshots and replication.

Network Virtualization: Virtual network allow admins to configure connectivity and isolate workloads without making every change directly on physical network devices.

Benefits of SDDC:

Automation - Reduces repetitive manual tasks

Agility - Enables faster provisioning of resources.

Scalability - Resources can be allocated according to demand. and available capacity.

Centralized management: Simplifies monitoring and administration.

Disaster recovery: Snapshots and replication can support recovery when properly configured.

5. Multi-cloud Federation failure due to identity mismatch:

Multi-cloud federation allows organizations to access services across multiple cloud providers. Identity mismatches occur when providers use different identity formats, authentication methods or authorization policies.

Causes of identity mismatches:

* different username and user identifier
* Incompatible authentication protocols
* Inconsistent roles and permissions.
* Different security policies.
* Privacy risks caused by excessive sharing of user information.

Proposed Secure Solution:

Use a federated Identity Provider:

A trusted IdP authenticates users and provides identity information to participating cloud providers.

Use Standard Federation Protocols:

* SAML 2.0: Exchanges authentication and identity assertions.
* OpenID Connect (OIDC): provides identity information through an authentication layer built on OAuth 2.0.
* OAuth 2.0: Enables delegated authorization to access protected resources.

Implement identity mapping:

Map users identities and roles consistently across providers. (Ex) a finance analyst should access only authorized financial resources while a student should access only permitted academic services.

Strengthen Security:

* Enable multi-factor authentication
* Apply role-based access control
* Use short-lived tokens where supported
* Revoke access when users leave the organization.

Protect Privacy

Encrypt communications, share only necessary user information and establish clear data access and retention policies.

Enable monitoring:

Maintain authentication logs, monitor suspicious activity and regularity review permission.

workflow:

